DSP Quiz # 1- Summer 26- All Laws

Date: 19 / 07 / 2026 & Time: 20 m (3 Marks)

Q # 1- Solve the following parts:

a- In sampling process, derive the sampling condition, $Fs \geq 2fmax$. **2**

b- In detail, show the main characteristics of the continuous time sinusoidal signals? **3**

Q # 2- For the following signal: $x_a(t) = A \cos(2\pi f_0 t)$, where, A=2 volt and f_0 =10 KHz.

a- Check the periodicity of this signal and find the fundamental period. **2**

b- If this signal is sampled at F_s= 5 KHz, is it possible to reconstruct the original signal from its samples? **3**

c- What is the minimum sampling frequency in this case? **2**

d- Using the minimum sampling frequency obtained par (c), find x(n) and calculate the sample values during one cycle of this signal and then sketch both analog and discrete time signals on one graph. **2 + 2 + 2**

Q # 3- Check (✓) for the right statements and (X) for the wrong statements and correct the mistakes. **3**

a- In quantization process, when the quantization step (Δ) increases, the accuracy of the quantizer increases. ( )

b- Nyquist frequency (F_N), is the minimum sampling frequency ( )

c- Fourier transform (FT) is calculated for a non-periodic continuous – time signals. ( )

GOOD LUC